Dataset Report — NASA POWER Weather

Master Thesis: Crop Harvesting Prediction · Winter wheat · Centre-Val de Loire, France

CLIMATE FEATURES

Dataset role: Weather/climate features · **Provider:** NASA POWER (Prediction Of Worldwide Energy Resources)
Access: NASA POWER REST API, Daily Point, community AG · **Endpoint:** power.larc.nasa.gov
License: Free and open (NASA Langley Research Center)

1. Source and provenance

1.1 What is NASA POWER?

NASA POWER provides global meteorological and solar data derived from NASA satellite observations and model reanalysis, designed for renewable-energy and agricultural applications. Its "AG" community endpoint is tailored to agronomy. Data are free, require no registration key, and are accessed through a simple REST API.

1.2 Why not satellite/GEE for weather?

Unlike vegetation indices, weather varies little between nearby parcels (it is a large-scale phenomenon). NASA POWER provides ready-to-use daily values by coordinate, so weather did not require Google Earth Engine or per-parcel satellite processing — a simpler, fully reproducible path.

1.3 Spatial and temporal scope

Item	Detail
Spatial sampling	6 points — one per department of Centre-Val de Loire
Points (préfectures)	Chartres (28), Orléans (45), Blois (41), Tours (37), Bourges (18), Châteauroux (36)
Temporal range	1 Oct 2019 → 31 Jul 2024 (1,766 days per point)
Why this range	Covers the 5 winter-wheat campaigns (sown Oct N-1, harvested Jul N)
Total raw rows	10,596 (6 points × 1,766 days)

2. Raw variables retrieved

Variable	Unit	Description
`T2M_MAX`	°C	Daily maximum air temperature at 2 m
`T2M_MIN`	°C	Daily minimum air temperature at 2 m
`T2M`	°C	Daily mean air temperature at 2 m
`T2MDEW`	°C	Dew point temperature at 2 m
`PRECTOTCORR`	mm/day	Bias-corrected total precipitation
`RH2M`	%	Relative humidity at 2 m
`ALLSKY_SFC_SW_DWN`	MJ/m²/day	Surface solar radiation (all-sky)
`WS2M`	m/s	Wind speed at 2 m

3. Transformations applied

Step 1 — API retrieval. Queried the NASA POWER Daily Point API once per department (6 requests), parsed the JSON response into a table, and merged the six points.

Step 2 — Campaign labelling. Assigned each day to an agricultural campaign (Oct N-1 → Jul N), identified by the harvest year, so cumulative features reset each season.

Step 3 — Thermal features. Computed Growing Degree Days at base 0 °C and base 5 °C, plus their per-campaign cumulatives; thermal amplitude (Tmax − Tmin); hot-day flag (Tmax > 25 °C) and cumulative hot-day count.

Step 4 — Reference evapotranspiration. Computed ET0 with the Penman-Monteith FAO-56 method from temperature, wind, solar radiation and humidity.

Step 5 — Water-balance features. Derived daily and cumulative water balance (precipitation − ET0), cumulative ET0, cumulative precipitation, and a 30-day rolling rainfall sum.

Step 6 — Energy feature. Computed cumulative solar radiation per campaign.

Step 7 — Export. Organised the 30 columns and saved the final CSV.

4. Key decisions and rationale

Decision: Six weather points (one per department) rather than one per parcel.
Rationale: Weather is spatially smooth — parcels 10–20 km apart see nearly identical conditions. Six points capture the region's north–south gradient while avoiding redundant queries. Each parcel is later linked to the weather of its department.

Decision: Compute Growing Degree Days (base 0 and base 5).
Rationale: Wheat develops with accumulated heat, not the calendar — GDD is the standard agronomic driver of phenology. Both bases are provided; base 0 is standard for French winter wheat.

Decision: Use Penman-Monteith FAO-56 for ET0.
Rationale: The FAO-56 method is the international scientific standard, enabling a true water-balance feature, more rigorous than temperature-only approximations.

Decision: Engineer cumulative, campaign-based features.
Rationale: The harvest is driven by accumulated conditions over the whole season; cumulative features that reset each campaign capture this progression directly.

5. Quality assurance

5.1 Raw data checks

- 0 missing values across all 10,596 rows.
- 0 NASA POWER fill-codes (−999) in any variable.
- Values within realistic physical ranges for the region.

5.2 A corrected error (transparency)

Issue found and fixed: the first ET0 implementation mistakenly converted solar radiation by a ×3.6 factor, producing impossible values (up to ~24 mm/day in July). Sanity-checking against expected ranges revealed the problem: NASA POWER already delivers radiation in MJ/m²/day, so no conversion was needed. After correction, ET0 is realistic — about 0.2–1 mm/day in winter and 4–6 mm/day in July (mean 2.14 mm/day). This illustrates the validation discipline applied throughout the project.

6. Resulting deliverable

Item	Value
Dimensions	10,596 rows × 30 columns
Identifiers	dept, ville, lat, lon, date, saison, doy
Raw variables	8 (temperature, precipitation, humidity, radiation, wind, dew point)
Thermal features	GDD (base 0 & 5) + cumulatives, thermal amplitude, hot-day counts
Water features	ET0 + cumulative, precipitation + cumulative, water balance + cumulative, 30-day rolling rain
Energy feature	Cumulative solar radiation
Output file	`meteo_nasapower_cvl_2020_2024.csv`
Documentation	`Data_Dictionary_Meteo_NASAPOWER.pdf` (3 pages)

7. Relevant observations

- **Agronomic signal already visible:** in 2023, the Chartres point reached GDD5_cumul ≈ 2,000 with a strongly negative cumulative water balance and falling 30-day rainfall by early June — conditions consistent with the observed early harvest (3 July 2023).
- **Expected most-predictive features:** cumulative GDD (development stage), cumulative water balance (stress), cumulative hot days (ripening acceleration), and 30-day rolling rain (pre-harvest grain moisture).
- **Join strategy:** each parcel will be linked to weather by its department and date (or an aggregated 10/15-day window) during master-dataset assembly.

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